

IRA Apprenticeship Compliance Dashboard

Monitoring workforce ratios, labor hours, and compliance performance across large-scale solar projects

Project Overview

I developed this dashboard to support workforce compliance monitoring for utility-scale solar projects operating under Inflation Reduction Act apprenticeship requirements. The goal was to give project, delivery, and leadership teams a reliable way to monitor apprenticeship labor thresholds, apprentice-to-journeyworker ratios, labor hours, and compliance exceptions across active jobs and reporting periods.

At the site level, compliance was governed by daily apprentice-to-journeyworker ratio requirements by craft. If apprentices outnumbered journeyworkers within a craft on a given day, certain hours could be disqualified from counting toward the program, creating potential financial exposure and jeopardizing higher tax credit eligibility.

The dashboard translated complex workforce activity into a clear reporting framework that allowed users to monitor overall compliance status, identify ratio risks by project and craft, and validate supporting employee-level detail when needed. Rather than relying on disconnected spreadsheets or reactive reporting, the solution centralized workforce counts, labor hours, classifications, pay codes, and exceptions into one operational view.

This work supported a broader renewable energy compliance initiative in which our largest client exceeded the required apprenticeship threshold by reaching more than 15% apprenticeship hours, preserved eligibility for significant tax credit benefits across all our clients and projects, and established a repeatable model for future IRA-regulated projects.

Business Challenge

IRA labor compliance created both operational and financial risk. Project teams had to coordinate large numbers of workers across multiple jobsites, crafts, and reporting periods while maintaining required apprenticeship participation levels. As workforce movement increased, it became harder to identify ratio issues quickly enough to prevent downstream compliance problems.

From an analytics standpoint, the reporting needed to answer four key questions:

- Are we currently meeting apprenticeship participation requirements?
- Which project, craft, or date is creating ratio risk?
- How much apprentice labor is contributing to total work hours?
- Can users validate the numbers at the employee and pay-code level?

Solution & Dashboard Design

I structured the dashboard around the way different users needed to make decisions: leadership needed summary visibility, while operations and payroll teams needed the ability to investigate exceptions and validate details.

Executive Summary View

The summary layer provides a high-level view of apprentice counts, journeyworker counts, ratio status, apprentice hours, journeyworker hours, total hours, and apprentice-hour percentage. This gives leaders a fast read on whether the project is tracking toward compliance during the reporting period.

Craft-Level Performance Analysis

The dashboard breaks performance down by craft, such as Construction Craft Laborer and Equipment Operator. This is important because overall project performance can appear compliant while a specific craft is creating ratio risk. The craft-level view helps isolate where corrective action may be needed.

Exception & Compliance Risk Monitoring

The ratio error view identifies projects, dates, and instances where apprentice-to-journeyworker ratios fall outside the required threshold. This allows operations teams to focus on the specific areas creating compliance exposure instead of manually reviewing every project and craft.

Employee-Level Detail & Auditability

The employee detail page provides the supporting audit trail behind the summary metrics. It includes employee ID, classification, pay code, daily hours, and total hours, allowing users to validate calculations, reconcile payroll activity, and understand which records are contributing to the final compliance results.

Data Development & Cross-Functional Collaboration

In addition to developing the dashboard, I partnered with cross-functional teams to support the data source powering the report. This included helping define and validate business logic for apprentice versus journeyworker classification, hours aggregation, ratio calculations, pay-code handling, and project-level reporting rules.

I also supported validation using SQL in Snowflake, comparing results across employee, project, and craft levels and reconciling aggregated metrics against source system outputs. Through this process, I helped identify discrepancies in classification logic and hour calculations and worked with upstream teams to resolve data quality issues before they impacted reporting accuracy.

This work helped strengthen the underlying data model so the reporting could support not only this dashboard, but additional workforce and compliance reporting needs across future projects.

Key Insights & Analytical Value

The dashboard moved compliance monitoring from a reactive process to a more proactive operating rhythm. Users could quickly identify where ratio issues were occurring, understand whether risk was isolated or recurring, and drill into the underlying labor records to support follow-up action.

The primary analytical value was the combination of summary visibility and exception-based detail. Leadership could assess overall compliance health, while operational teams could use the same report to investigate specific project, craft, date, or employee-level drivers.

Business Impact & Outcomes

This dashboard supported a broader program that delivered measurable outcomes:

- Exceeded the required apprenticeship threshold by achieving more than 15% total apprenticeship hours
- Helped preserve eligibility for significant tax credit benefits tied to IRA labor compliance
- Established a reporting standard for current and future IRA-regulated projects
- Supported compliance monitoring across a workforce environment spanning multiple clients, projects, workers, and labor-hour volumes
- Reduced reliance on disconnected spreadsheets and manual reporting workflows

From an analyst perspective, the value extended beyond the dashboard itself. The project required translating regulatory requirements into measurable KPIs, validating the data behind those KPIs, and designing a reporting experience that supported real operational decisions.

Tools & Skills Demonstrated

- Tableau dashboard design for operational decision support
- Compliance and exception-based reporting
- Workforce hour and ratio analysis
- Multi-level drilldown design
- Translation of regulatory requirements into measurable KPIs
- SQL-based validation in Snowflake
- Business logic standardization across reporting layers
- Cross-functional collaboration with data, payroll, workforce, and operations teams
- Data quality review and reconciliation
- Executive summary reporting with employee-level audit support

Project Relevance

This project reflects the type of analytics work that creates measurable value in complex, regulated operating environments. It required turning compliance requirements into clear metrics, building a reporting experience that supported both leadership visibility and operational follow-through, and helping ensure the underlying data was accurate enough to support high-stakes decisions.

For organizations managing large workforce populations, financial exposure, and regulatory obligations, this type of solution demonstrates how analytics can move beyond reporting and become a practical tool for risk management, process improvement, and enterprise decision support.